

Sentiment-to-Price Correlation Analysis Report

1. Executive Summary

This report analyzes the linear correlation between FinBERT-extracted news sentiment and next-day stock returns across 20 major NSE equities.

- Total Data Samples Analyzed: 2583 trading days
- Overall Pearson Correlation (\$r\$): 0.0337
- Statistical Significance (\$p\$-value): 0.0865
- Overall Assessment: Weak / Moderate Correlation

2. Correlation Breakdown by NSE Stock

Symbol	Pearson \$r\$	\$p\$-value	Significant (p < 0.05)	Sample Size
WAAREEENER	0.1935	0.0320	Yes (p < 0.05)	123
WIPRO	0.1681	0.0630	No	123
AXISBANK	0.1056	0.2453	No	123
ITC	0.0887	0.3295	No	123
LT	0.0822	0.3659	No	123
SUNPHARMA	0.0793	0.3833	No	123
TCS	0.0791	0.3844	No	123
RELIANCE	0.0437	0.6313	No	123
HDFCBANK	0.0370	0.6843	No	123
BHARTIARTL	0.0340	0.7092	No	123
BAJFINANCE	0.0259	0.7765	No	123
ASIANPAINT	0.0090	0.9217	No	123
TITAN	-0.0125	0.8905	No	123
INFY	-0.0211	0.8172	No	123
NTPC	-0.0306	0.7369	No	123
ICICIBANK	-0.0310	0.7339	No	123
MARUTI	-0.0320	0.7251	No	123
HCLTECH	-0.0554	0.5429	No	123
SBIN	-0.0993	0.2745	No	123
HINDUNILVR	-0.1043	0.2510	No	123
KOTAKBANK	-0.1180	0.1938	No	123

3. Key Observations & Findings

1. Stocks with strong positive sentiment-return alignment display predictive signals suitable for reinforcement learning trading state representation.
2. News volume and sentiment variance serve as important indicators of price volatility on subsequent trading days.
3. Lagged features (1-day and 2-day prior sentiment) often carry stronger predictive signal than same-day sentiment due to news dissemination delays.